

Wazuh SIEM Home Lab

Cybersecurity Portfolio Project

Sean Murphy | April 2026

Author	Sean Murphy
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Platform	VirtualBox 7.2.6 on Windows 11
OS	Ubuntu 24.04.4 LTS
SIEM	Wazuh 4.7.5 (All-in-One)
Monitored Agent	Windows 11 Laptop (via NAT Port Forwarding)
Certifications	CompTIA Security+

1. Executive Summary

This project documents the end-to-end deployment of a Wazuh Security Information and Event Management (SIEM) system in a home lab environment. The objective was to build hands-on experience with enterprise-grade security monitoring tools, endpoint agent deployment, and real-time threat detection — skills directly applicable to SOC Analyst, Cybersecurity Analyst, and IT Security roles.

The final result is a fully operational SIEM that monitors a live Windows 11 endpoint, detecting authentication events, file integrity changes, policy violations, and MITRE ATT&CK framework-mapped threats in real time.

2. Lab Environment

Hardware

- Host Machine: HP Laptop, Intel CPU, 16GB RAM, 366GB free disk space
- Operating System: Windows 11
- Virtualization: Oracle VirtualBox 7.2.6

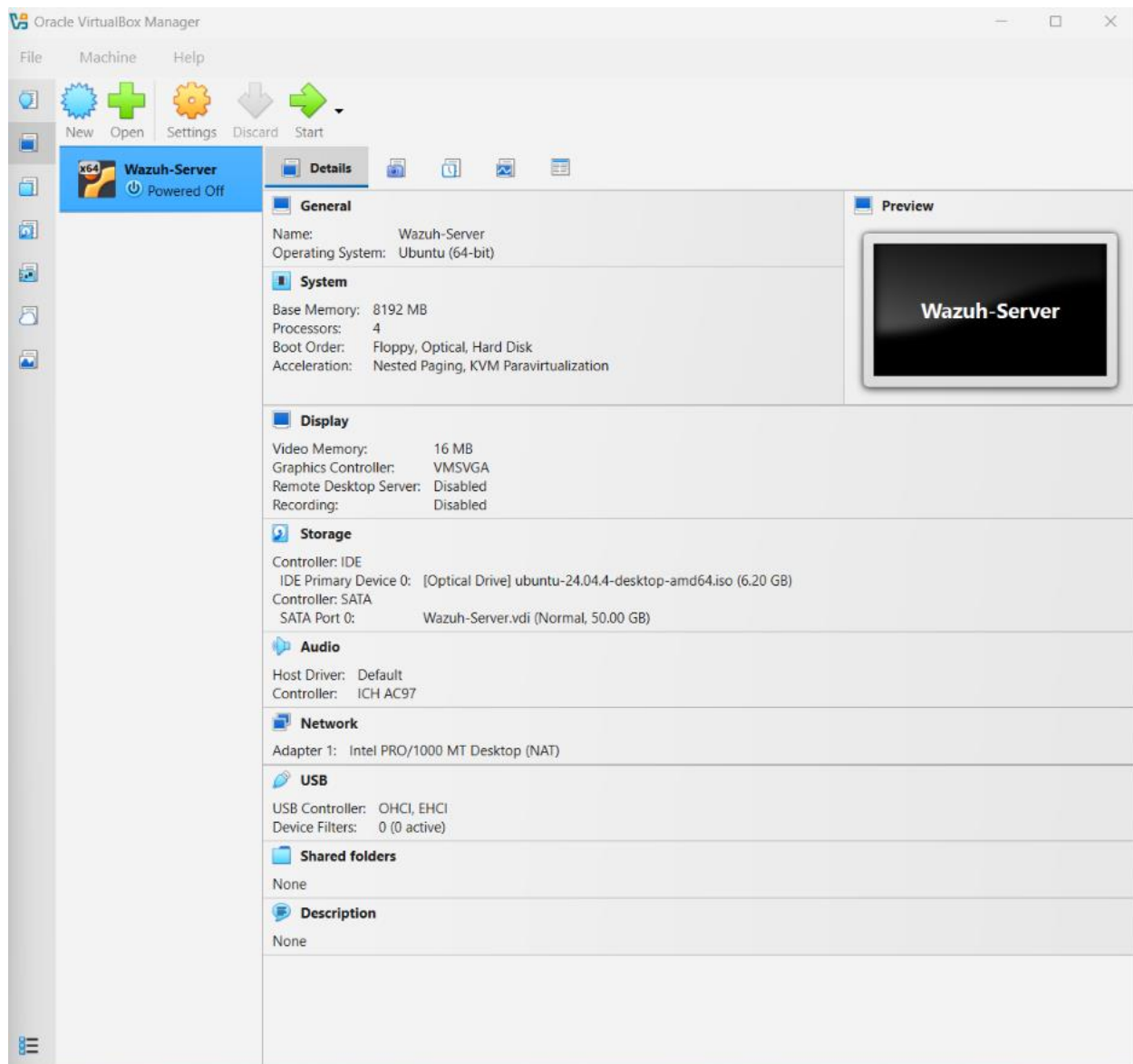
Virtual Machine Specifications

- VM Name: Wazuh-Server
- OS: Ubuntu 24.04.4 LTS (Desktop)
- RAM Allocated: 8GB (8192 MB)
- CPU Cores: 4

- Disk: 50GB dynamically allocated VDI
- Network: NAT with port forwarding rules

Software Stack

- Wazuh Manager 4.7.5
- Wazuh Indexer (OpenSearch-based)
- Wazuh Dashboard
- Filebeat
- Wazuh Windows Agent 4.7.5



3. Installation Process

Step 1: VM Creation

Created a new VirtualBox virtual machine named Wazuh-Server, configured with 8GB RAM and 4 CPU cores to meet Wazuh's resource requirements. Attached the Ubuntu 24.04.4 desktop ISO and provisioned a 50GB virtual hard disk.

Step 2: Ubuntu Installation

Performed an interactive Ubuntu installation, selecting default disk partitioning and creating a local user account. Post-installation, ran a full system update to ensure all packages were current before proceeding.

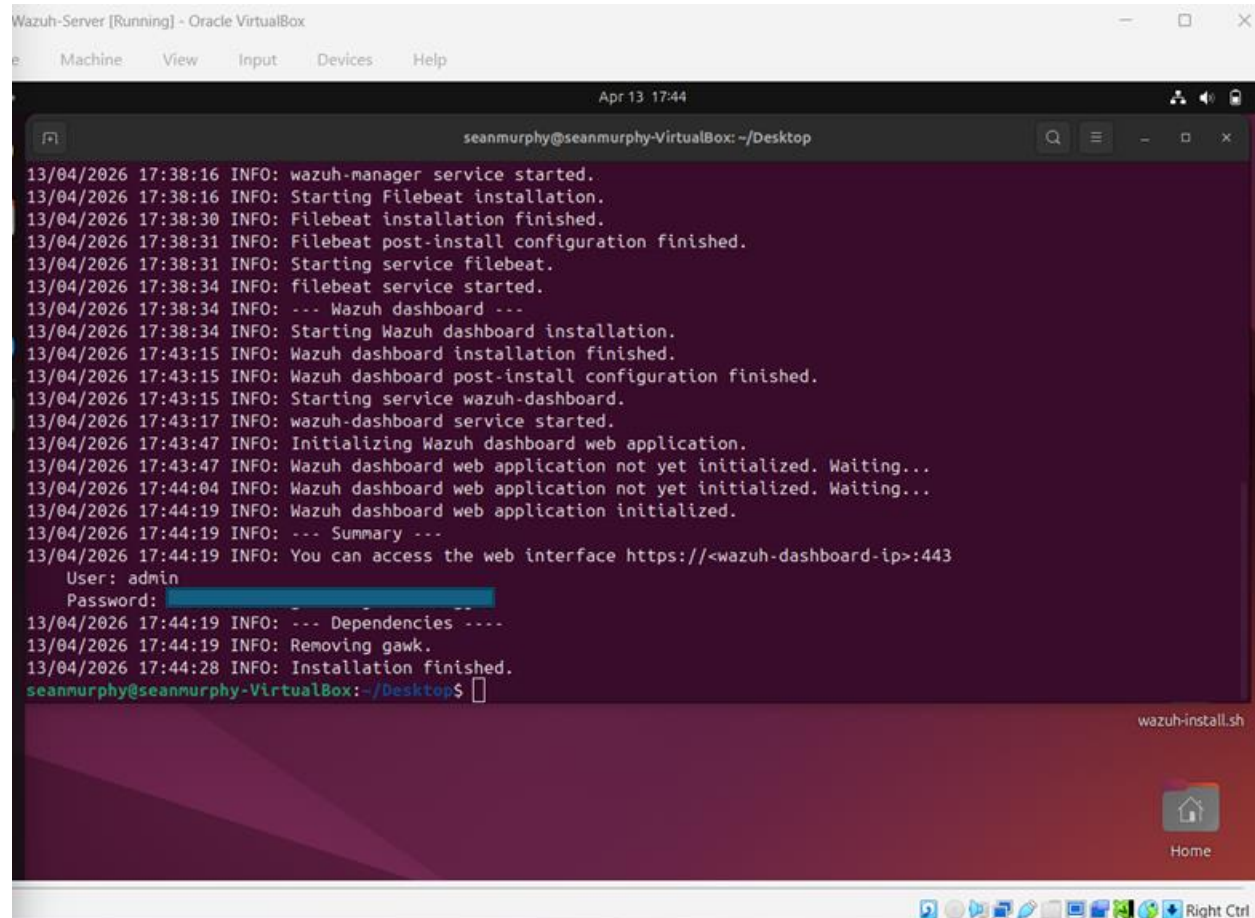
Command used: `sudo apt update && sudo apt upgrade -y`

Step 3: Wazuh All-in-One Installation

Used Wazuh's official quick install script to deploy the complete stack (manager, indexer, and dashboard) on a single VM. The `-i` flag was required to bypass an OS compatibility check since Wazuh 4.7.5 does not yet officially list Ubuntu 24.04 — however, the installation and all services functioned correctly.

Command used: `curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh && sudo bash wazuh-install.sh -a -i`

Installation time: approximately 15 minutes. Upon completion, the installer output admin credentials for dashboard access.



```
Wazuh-Server [Running] - Oracle VirtualBox
e Machine View Input Devices Help

Apr 13 17:44
seanmurphy@seanmurphy-VirtualBox: ~/Desktop

13/04/2026 17:38:16 INFO: wazuh-manager service started.
13/04/2026 17:38:16 INFO: Starting Filebeat installation.
13/04/2026 17:38:30 INFO: Filebeat installation finished.
13/04/2026 17:38:31 INFO: Filebeat post-install configuration finished.
13/04/2026 17:38:31 INFO: Starting service filebeat.
13/04/2026 17:38:34 INFO: filebeat service started.
13/04/2026 17:38:34 INFO: --- Wazuh dashboard ---
13/04/2026 17:38:34 INFO: Starting Wazuh dashboard installation.
13/04/2026 17:43:15 INFO: Wazuh dashboard installation finished.
13/04/2026 17:43:15 INFO: Wazuh dashboard post-install configuration finished.
13/04/2026 17:43:15 INFO: Starting service wazuh-dashboard.
13/04/2026 17:43:17 INFO: wazuh-dashboard service started.
13/04/2026 17:43:47 INFO: Initializing Wazuh dashboard web application.
13/04/2026 17:43:47 INFO: Wazuh dashboard web application not yet initialized. Waiting...
13/04/2026 17:44:04 INFO: Wazuh dashboard web application not yet initialized. Waiting...
13/04/2026 17:44:19 INFO: Wazuh dashboard web application initialized.
13/04/2026 17:44:19 INFO: --- Summary ---
13/04/2026 17:44:19 INFO: You can access the web interface https://<wazuh-dashboard-ip>:443
User: admin
Password: [redacted]
13/04/2026 17:44:19 INFO: --- Dependencies ---
13/04/2026 17:44:19 INFO: Removing gawk.
13/04/2026 17:44:28 INFO: Installation finished.
seanmurphy@seanmurphy-VirtualBox: ~/Desktop$
```

4. Troubleshooting & Problem Solving

A key learning outcome of this project was working through real technical obstacles — the kind encountered in production environments.

Issue 1: OS Compatibility Check Failure

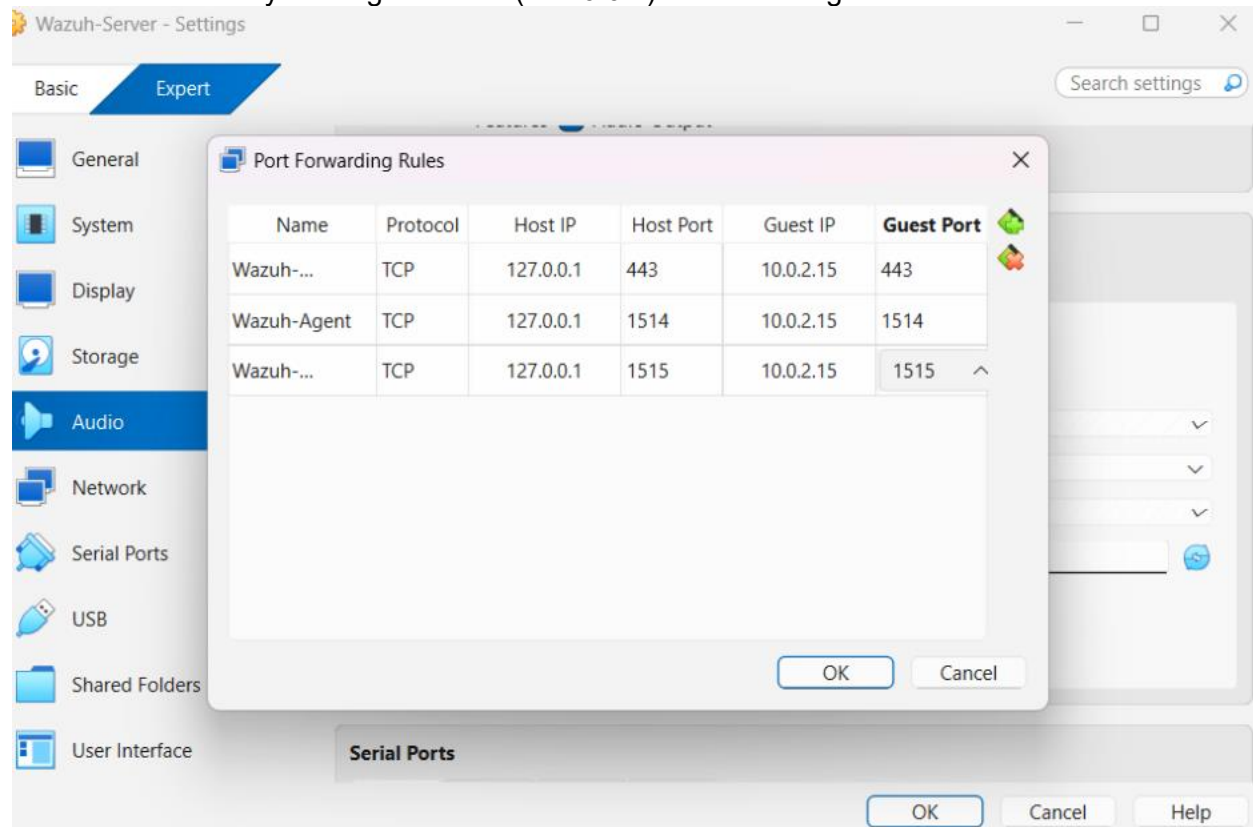
Problem: Wazuh installer rejected Ubuntu 24.04 as unsupported.

Solution: Used the `-i` (ignore-check) flag to bypass the version check. All services were deployed and started successfully.

Issue 2: VirtualBox Host-Only Networking Driver Failure

Problem: VirtualBox 7.2.6 failed to create a Host-Only network adapter due to a missing driver (VERR_FILE_NOT_FOUND). Bridged networking was also unavailable due to the same underlying driver issue.

Solution: Pivoted to NAT networking with custom port forwarding rules on ports 443, 1514, and 1515. This allowed the Windows agent to communicate with the Wazuh server through VirtualBox's NAT layer using localhost (127.0.0.1) as the manager address.



Issue 3: Package Conflict (wazuh-manager vs wazuh-agent)

Problem: Attempting to install wazuh-agent via apt caused wazuh-manager to be automatically removed due to a package conflict.

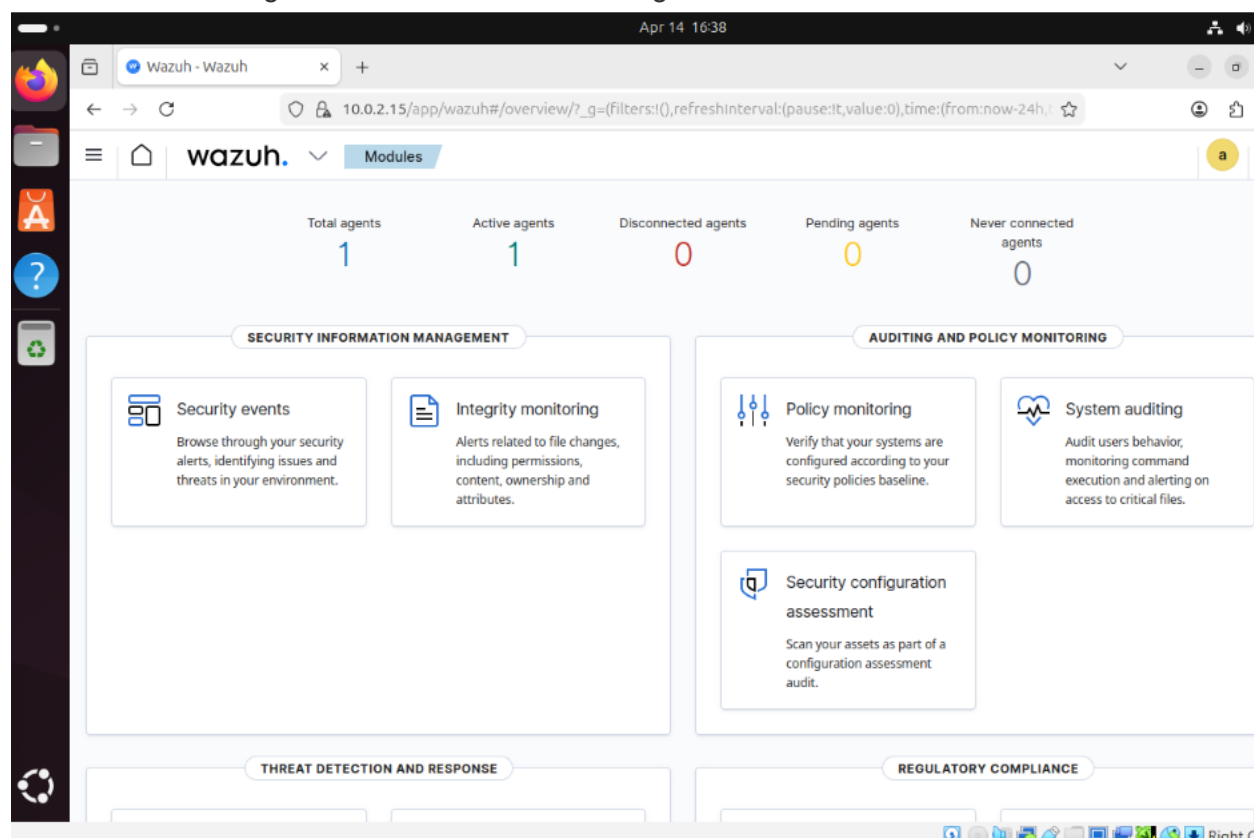
Solution: Reinstalled the full Wazuh stack using the --overwrite flag. Deployed the agent on the Windows host instead, which avoids the package conflict entirely and is a more realistic production architecture.

5. Windows Agent Deployment

Installed the Wazuh Windows agent (MSI package) on the host Windows 11 machine using PowerShell with administrator privileges. Configured the agent to point to 127.0.0.1 as the manager address, leveraging VirtualBox's NAT port forwarding to route traffic to the VM.

Install command: `msiexec.exe /i wazuh-agent.msi /q WAZUH_MANAGER='127.0.0.1' WAZUH_REGISTRATION_SERVER='127.0.0.1'`

After starting the agent service (NET START WazuhSvc), the Wazuh dashboard immediately reflected 1 Active Agent with real-time data flowing in.



6. Results & Key Findings

Within minutes of agent connection, the Wazuh dashboard populated with real security telemetry from the Windows endpoint:

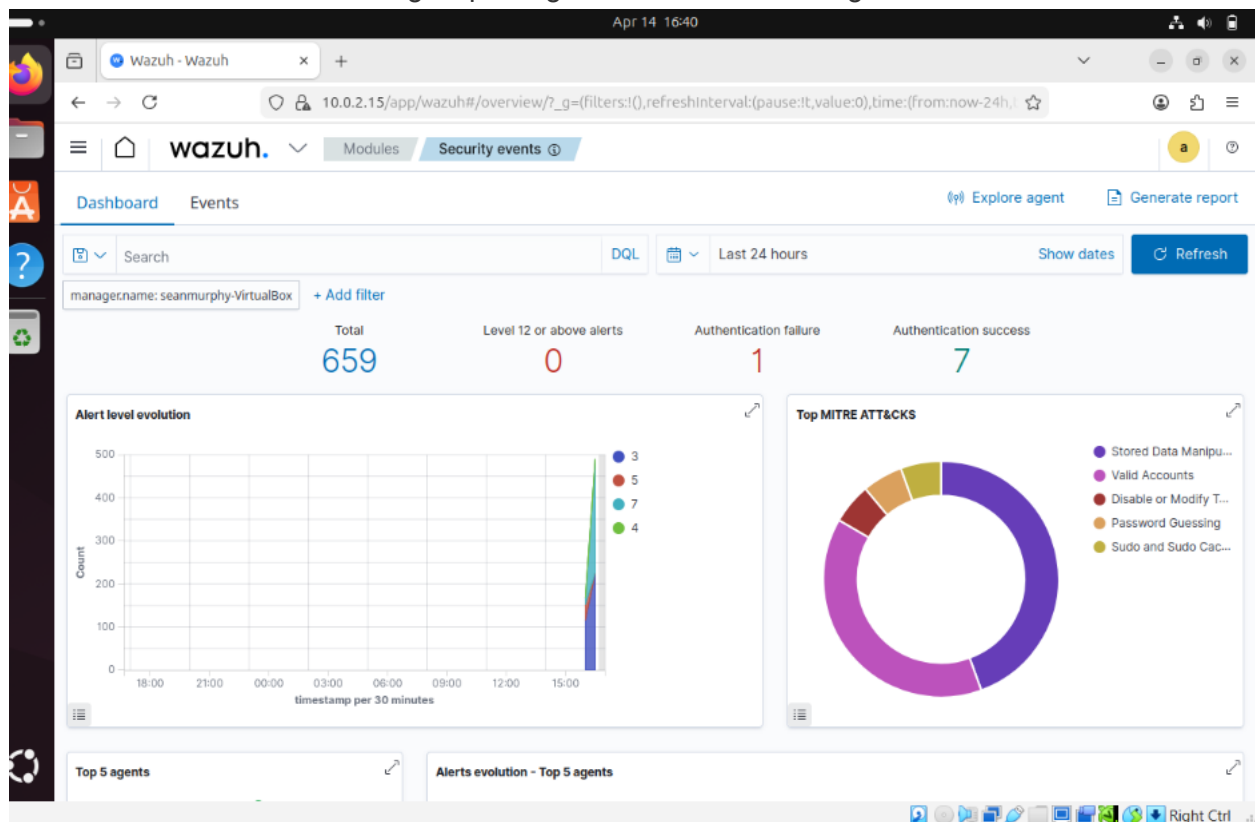
Metric	Value
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Total Security Events	659
Critical Alerts (Level 12+)	0
Authentication Failures	1
Authentication Successes	7
MITRE ATT&CK Categories Detected	5
Active Monitored Agents	1

MITRE ATT&CK Detections

Wazuh automatically mapped detected events to the MITRE ATT&CK framework, identifying the following technique categories:

- Stored Data Manipulation — activity involving stored files or system data
- Valid Accounts — use of legitimate credentials for authentication
- Disable or Modify Tools — attempts to alter security tooling
- Password Guessing — failed authentication attempts
- Sudo and Sudo Caching — privilege escalation monitoring



7. Skills Demonstrated

- Linux system administration (Ubuntu 24.04, apt package management, systemctl)
- Virtualization and VM configuration (VirtualBox, NAT networking, port forwarding)
- SIEM deployment and administration (Wazuh all-in-one architecture)
- Endpoint agent deployment (Windows MSI installation via PowerShell)
- Network troubleshooting (NAT vs Bridged vs Host-Only adapter diagnosis)
- Security event analysis (alert severity levels, authentication monitoring)
- MITRE ATT&CK framework familiarity
- Technical problem solving and documentation

8. Next Steps

Future iterations of this lab will include:

- Simulating attack scenarios using Kali Linux to generate detectable threats
- Building custom Wazuh detection rules for specific threat patterns
- Exploring Wazuh's File Integrity Monitoring (FIM) module
- Adding an Active Directory lab to simulate an enterprise environment
- Writing incident response reports based on Wazuh alert data